# Shubhadeep Mandal

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## Education

**International Institute of Information Technology, Hyderabad** July 2025 – Present
*Telangana, India*
*Master of Technology in Computer Science and Engineering* *CGPA: 9.12*

**Indian Institute of Information Technology, Ranchi** Aug 2020 – Jun 2024
*Ranchi, India*
*Bachelor of Technology (Hons.) in Computer Science and Engineering* *CGPA: 9.05*

## Projects

**BoardBuddy: Real-Time Collaborative Whiteboard** | *Next.js 14, Convex DB, Liveblocks, TypeScript*

- Developed a full-stack collaborative canvas supporting multi-user synchronization, freehand sketching, geometric shapes, and undo/redo.
- Synchronized real-time state and user presence across sessions using Liveblocks WebSockets and Convex reactive database.
- Secured organization-scoped workspaces and role permissions via Clerk authentication and responsive TailwindCSS/shadcn UI.

**AI-SaaS: Multi-Modal Generation Platform** | *Next.js 13, TypeScript, Prisma, Stripe, PostgreSQL*

- Architected a multi-modal SaaS platform integrating OpenAI and Replicate APIs for conversational AI, code, audio, and video synthesis.
- Engineered customer subscription billing, tiered usage quotas, and automated webhook event handlers using Stripe SDK and Clerk.
- Optimized relational database queries and schema migrations with Prisma ORM deployed over managed PostgreSQL.

**Real-Time Social Messaging Platform** | *Next.js, TypeScript, Pusher Channels, Prisma, MongoDB*

- Engineered instant 1-on-1 and group chat rooms with real-time WebSocket event broadcasting powered by Pusher Channels.
- Implemented user online/offline presence tracking, active typing indicators, image uploads, and per-user message read receipts.

**Peer-to-Peer Distributed File Sharing System** | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Engineered a BitTorrent-style hybrid P2P system in C++, achieving an **8.5x transfer speedup on 1GB files** (193s to 22.8s) across 5 concurrent peers by saturating network and disk I/O.
- Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads.
- Architected Primary-Backup tracker failover with 2s bounded failover, full state synchronization, and verified resilience via **48 unit tests** under AddressSanitizer/UBSan.

## Technical Skills

**Languages**: TypeScript, JavaScript, Python, C/C++, SQL, HTML/CSS, Bash.
**Frontend Engineering**: Next.js 14, React, TailwindCSS, shadcn/ui, Redux, HTML5 Canvas API, WebRTC.
**Backend & Cloud**: Node.js, Convex, Prisma ORM, PostgreSQL, MongoDB, REST APIs, Stripe SDK, Clerk.
**Real-Time & Tools**: WebSockets (Liveblocks, Pusher), Sockets, Linux/Unix, Git, Docker, Vercel.

## Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

## Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.